PROJECT ID: CINT-11 · CAPSTONE REVIEW II

Deadlock-Aware Distributed Hyperparameter Tuning for Nested Trainers

NestedTune Guard • Deterministic Simulation • Hold-and-Wait Prevention • Production Admission Control

Abstract

OVERVIEW

Distributed hyperparameter optimization (HPO) frameworks like Ray Tune launch nested training workers (e.g. XGBoost/PyTorch DDP) requiring multi-tier placement groups (PGs). When outer trials reserve parent PGs and nested workers request child PGs without coordinated visibility, circular dependencies induce catastrophic **hold-and-wait deadlocks (Ray Issue #47439)**, stranding cluster capacity indefinitely.

NestedTune Guard establishes a rigorous discrete-event simulator and mathematical evaluation framework paired with an atomic, deadlock-aware admission controller. By verifying combined parent-plus-child feasibility before admitting trials and rejecting permanently infeasible requests, our system completely eliminates circular wait states (0 cycles vs. 4 cycles in baselines) while preserving maximal cluster completion throughput.

Distributed HPO | Ray Tune / Train | Placement Groups | Hold-and-Wait Deadlock | Admission Control | Discrete-Event Simulation

Problem Statement: Ray Issue #47439

ANOMALY

In multi-tier training workflows, hierarchy-blind schedulers (FIFO & Greedy) admit trials based solely on outer parent placement group requests:

> **The 4/4/1/3 Accounting Deadlock Pattern:**
> • Cluster Capacity: **4 CPUs**.
> • Parent Trial reserves: **4 CPUs** (Outer PG granted).
> • Actually useful worker execution: **1 CPU**.
> • Stranded capacity held by parent: **3 CPUs**.
> • Nested Child Trainer requests: **3 CPUs**.
> • **Failure:** Child cannot obtain 3 CPUs because parent holds them; parent cannot release CPUs until child finishes.

- **Self-Cycle Deadlock:** Trial enters indefinite `WAITING_CHILD` state. Wait-for graph exhibits circular dependency: `xgb-001 → xgb-001`.
- **Capacity Stranding:** 75% of reserved cluster compute is locked idle, preventing any subsequent queue progression.
- **Silent Cost Inflation:** Cloud instances run indefinitely without progress, causing runaway cloud bills and aborted tuning sweeps.

Research Objectives

GOALS

- **Real-Ray Reproduction:** Validate the deadlock inside isolated Docker containers running real Ray 2.10.0, matching exact empirical accounting.
- **Deterministic Modeling:** Design an event-driven simulator with a 30s bounded horizon, providing reproducible baselines without timing jitter.
- **Deadlock-Aware Admission Control:** Enforce combined parent+child feasibility invariance prior to granting any cluster resources.
- **Explainable Diagnostic Advisor:** Build a pure diagnostic rule engine classifying trial stalls into capacity shortfalls vs. deadlock hazards.
- **Full-Stack Tooling & Dashboard:** Express FastAPI REST microservices and a Next.js 14 reactive